

Lab #6: Hardware Implementation of a 4-Bit Calculator Using Full-Adder Chips

In this lab, you will build a 4-bit adder-subtractor in hardware using the full adder and XOR chips. **This lab is to be done individually.**

Part 1: Designing the Full Adder-Subtractor

For this part of the lab, you will design a 4-bit adder-subtractor based on your schematic from Lab 4, Part 3. This will be almost the same circuit build as that from Lab 5, with the difference being that you will be working with the full adder and XOR chip. As an example, Figure 1 illustrates the least significant bit (LSB) of your circuit build:

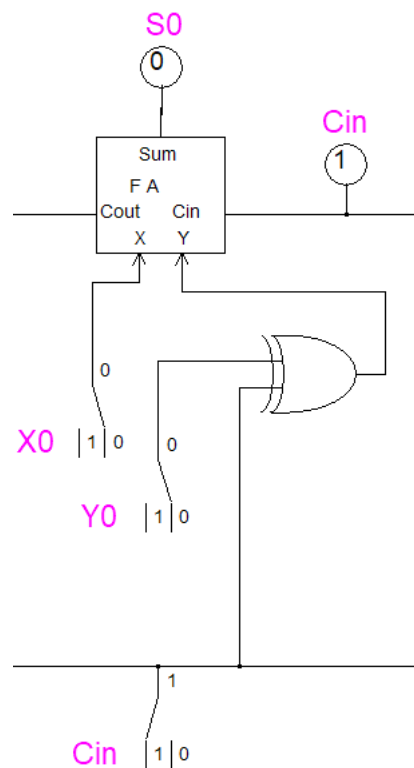


Figure 1: A Full Adder-Subtractor Cell (LSB)

Note: For Questions 2 and 3 on the next page, use the same values as those from Lab #5.

Questions:

1. Make your circuit schematic. Label all inputs and outputs appropriately.
2. Give **four** examples of $A + B$ **using your calculator**. (Hint: have $A > 0$ plus $B > 0$, $A > 0$ plus $B < 0$, $A < 0$ plus $B > 0$, and $A < 0$ plus $B < 0$). Place the inputs and results in a table and include *four* screenshots from your addition calculations. Caption your figures.
3. Give **four** examples of $A - B$ **using your calculator**. Make sure you place the inputs and results in a table and include *four* screenshots from your subtraction calculations. Caption your figures.

Part 2: Building your SimUAid Design in Hardware

You will now build your circuit design on your breadboard. For your hardware build, you are **only allowed to use the full adder and XOR** chips to complete your circuit. Follow the demo instructions. In your presentation, you should perform **3** addition and subtraction operations from the previous part's Question 2 and Question 3, respectively. Use LEDs as indicators for your calculator. Note the MSB and the LSB for X, Y, and S in your demo.

Questions:

4. What pull-up or pull-down resistors are you using and why?
5. Provide a demo link, or in-person demo note.

Demo notes:

Make sure that you put your name somewhere visible on your breadboard. Make sure you state your name in the video.

It is required that your video is continuous, i.e. there should be no pauses, edits or interruptions to the recording. However, you may speed it up as appropriate.

The demo must show you **DISASSEMBLING** your circuit in your video at the end of the demo. Failure to do so will result in a loss of 50% of your grade.

The demo must be no more than a 6-minute video. Add it to your Google Drive or YouTube and put a **publicly accessible** link to that video in your report.

Part 3: Summary of Deliverables

Turn in your report in either PDF or Word format, and make sure it is named:
ECE3441_StudentID#_Lab6.

****Note: Please refrain from using notebook paper or writing on the attached lab. Make a separate Word document and have it submitted as a word or as a pdf document in BB. Points will be deducted if you do not follow instructions.**

Use the following structure for your lab report:

Course Name and Number – Semester

Lab Number and Title

Name

Overview: Goals of the Lab

Briefly (~ 1 small paragraph) outline the goal of this lab, as well as the work done.

Step-by-Step Procedures and Results

This must be in chronological order of how Lab 6 should be completed. Make sure you include all the necessary information that is mentioned in the lab and answer all questions.

Conclusion

Summarize what you have learned in this lab.